# Rufat Asadli

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## Education

- **ETH Zurich [5.5/6.0]** *Sep'23-Oct'25*
  *MSc in Statistics*
- **Charles University [1.1/1.0]** *Sep'19-Jun'22*
  *BSc in Economics and Finance*
  - Recipient of **Merit Stipend for Outstanding Academic Achievements** for 2 years awarded to **top 10%** students across all departments.

## Experience

- **Networked Systems Group, ETH Zurich** *Oct'25 - Present*
  *Research Intern, Supervised by Prof. Dr. Laurent Vanbever* Zurich, Switzerland
  - Extending previous research on **benchmarking LLMs** for **code generation** and network **configuration synthesis** with novel **curriculum learning** algorithms.
- **NetFabric** *Mar'25 - Oct'25*
  *Graduate Student Researcher, Supervised by Prof. Dr. Laurent Vanbever and Dr. Benjamin Bichsel* Zurich, Switzerland
  - Developed the first robust **evaluation benchmark** of closed- and open-source **LLMs** for **constrained code generation** tasks via zero-shot learning and supervised fine-tuning experiments.
- **Rycolab, ETH Zurich** *Apr'24 - May'25*
  *Graduate Student Researcher, Supervised by Prof. Dr. Ryan Cotterell and Dr. Alex Warstadt* Zurich, Switzerland
  - Developed a **PPO**-based training setup where a **speaker language model (LM)** learns to communicate privileged info to a **listener LM**, receiving rewards based on the listener's comprehension accuracy.
- **Kapital Bank OJSC** *Oct'22 - Sep'25*
  *Data Science Researcher, AI Center of Excellence* Baku, Azerbaijan
  - Built a **high-frequency prediction model** using **ensemble methods** together with Fourier transforms for feature engineering, **calibrating** raw predictions with Monte Carlo sampling from error distribution.

## Projects

- **Towards Developmentally Plausible Rewards: Communicative Success as a Learning Signal for Interactive Language Models [Code][Paper]** *Apr'24-May'25*
  *Preprint*
  - We present an **interactive language model training** method inspired by child language acquisition, showing communication-based rewards signal grammaticality and shape speaker behavior.
- **Does Catastrophic Forgetting Happen in Tiny Subspaces? [Code] [Report]** *Oct'24-Jan'25*
  *For Deep Learning, ETH Zürich.* ***Grade: 6/6***
  - Explored **neural network subspace structure in continual learning** showing that task-specific model learning and forgetting primarily occurs in gradient subspaces with small eigenvalues.
- **TARMAC: Conversational AI Dispatcher [Website]** *Jul'24 - Dec'24*
  *Startup Project*
  - Contributed to a **conversational dispatch system** with private data **fine-tuning** and dynamic **chat buffer memory update**, learning to **convert user prompts into SQL queries** for truck data retrieval.

## Skills

- **Programming Languages:** Python, HTML, CSS
- **Packages and Softwares:** PyTorch, TensorFlow, Transformers, NLTK, LangChain, TRL, Unsloth, vLLM, Accelerate, NumPy, Git, Bash, Docker, SQL, LaTeX

## Awards and Achievements

- Recipient of State Program scholarship to study Master's degree abroad (Azerbaijan) *2023*
- TensorFlow Developer Certificate *2022*